3 Infrastructure

UNIT SPECIFICS

Through this unit we have discussed the following aspects:

- ***The present and future projection of various facets of Infrastructure development***
    - *Habitats - Megacities, Smart Cities, futuristic visions*
    - *Transportation - Roads, Railways & Metros, Tunnels (below ground, under water); Seaports, River ways and canals; Airfields and Airport, Futuristic systems*
    - *Energy generation - Hydro, Solar, Photovoltaic, Wind, Wave, Tidal, Geothermal, Thermal energy and New sources*
    - *Water resource management*
    - *Telecommunication - Towers, above-ground and underground cabling*
- ***Awareness of various Codes & Standards governing Infrastructure development***
    - *ISO Codes for infrastructure construction*
- ***Innovations and methodologies for ensuring Sustainability in Infrastructure***

Besides giving a large number of multiple choice questions as well as questions of short and long answer types marked in two categories following lower and higher order of Bloom's taxonomy, a list of references and suggested readings are given in the unit so that one can go through them for practice.

There is a "Know More" section, which has been carefully designed so that the supplementary information provided in this part becomes beneficial for the users of the book. It is important to note that for getting more information on various topics of interest some QR codes have been provided which can be scanned for relevant supportive knowledge This section mainly highlights applications of the subject matter for our day-to-day real life or/and industrial applications on variety of aspects, case study related to environmental, sustainability, social and ethical issues whichever applicable, and finally inquisitiveness and curiosity topics of the unit.

RATIONALE

This fundamental unit discusses the important facets of infrastructure for the better understanding of its practical aspects during practice of Civil engineering, in the context of Sustainable development and future trends. It further emphasizes the role of standards and codes, and highlights the innovations and methodologies the civil engineer can leverage to develop sustainable infrastructure.

UNIT OUTCOMES

List of outcomes of this unit is as follows:

U3-O1: Knowledge on the present and future projection of various facets of Infrastructure Development: Habitat, Transportation, Energy generation, Water provision and Telecommunication

U3-O2: Understanding/comprehension of various Codes & Standards governing Infrastructure development

U3-O3: Knowledge on innovations and methodologies for ensuring Sustainability in Infrastructure development

Unit-3 Outcomes	EXPECTED MAPPING WITH COURSE OUTCOMES (1- Weak Correlation; 2- Medium correlation; 3- Strong Correlation)						
	CO-1	CO-2	CO-3	CO-4	CO-5	CO-6	CO-7
U3-O1	3	3	3	-	3	2	1
U3-O2	1	1	1	2	1	1	3
U3-O3	2	1	2	1	2	3	2

The term 'infrastructure' can be traced to 1927 referring to the various systems, amenities and facilities of public works that enables the provision of services. For example, Transportation services is provided by the associated infrastructure of roads, rail lines, harbours, bridges, etc. and Irrigations and Water management service is provided via dams, canals, etc. Today, classified as 'hard' infrastructure, these physical systems support various other services such as, waste management, telecommunication, power generation.

Prud'homme (2004) enlisted the **common characteristics of infrastructure**, as follows:

(i) Infrastructure "*are capital goods*", i.e., it is a means to provide service and not consumed directly, requiring labour and other inputs for it to be useful.
(ii) It is "lumpy" and not incremental, i.e., it is of a limited capacity to handle demand and cannot meet growing demands. It also requires years for being built and and of being in use.
(iii) It is long lasting, having long term implications on maintenance and hence, financing.
(iv) It is space specific and immobile, having further implications on financial capital.
(v) Infrastructure and the service it renders is subject to market failures, decreasing costs, externalities, etc.
(vi) It is used by both, households and enterprises, as it increases welfare and productivity.

In contrast, 'soft' infrastructure refers to the institutions that maintain health, socio-cultural and economic standards of a society, such as, healthcare and financial facilities, law enforcement and education. Of these, some are 'critical' infrastructure, identified as per the priority of the Nation. For India, these are Power, Telecom, Aviation, Energy, Banking, Cybersecurity and Disaster Management.

Another classification of infrastructure is based on its direct or indirect impact on production. The types that are essential for either improving or impeding production and distribution, such as, power, irrigation, transport, communication, etc., are classified as 'Economic Infrastructure, while those that aid economic progress, such as, Education, Healthcare and Housing, are classified as 'Social Infrastructure'.

3.1. HABITATS

Together, these various facets of infrastructure respond to the physical characteristics of a region to support the use of resources and productivity by the inhabitants beyond the boundaries of their natural habitat, with the underlying motivation for socio-economic development. In the following Sections, the present and future projection of various these various facets are discussed.

European Nature Information System (EUNIS) defines a **habitat** as "*a place where plants or animals live normally, characterized primarily by its physical features – topography, plant or animal physiognomy, soil characteristics, climate, water quality, etc., and secondarily by the species of plants and animals that live there*". In 2004, EUNIS classified various habitats, such as, marine, coastal, inland surface water, grasslands, woodlands and forests, along with recognising habitats created out of human intervention, such as, cultivated agricultural, horticultural and domestic habitats, and constructed, industrial and other artificial habitats, such as cities.

3.1.1 Megacities

In 2018 "*World Urbanisation Prospects*" the UN identified very large cities with population of over 10million as '**Megacities**' and UN further enlisted 33 megacities of which half of these urban agglomerations are in India and China and 27 are in the developing regions of the world, termed as the 'global south'. Globally, this is projected to rise to 43 by 2030, and the number of cities with 1 to 5 million inhabitants is projected to grow to 597. The characteristics of a Megacity are; *size, rate of growth and complexity* in terms of administration and infrastructure (Wenzel et al., 2007), and are the economic drivers of the country.

The urban population is projected to increase up to 28% worldwide, while rural population is projected to fall to 40% by 2030 (UN, 2018). Tokyo, Delhi and Shanghai that ranked 1st, 2nd , 3rd back in 2018 has already been overtaken by Shanghai, Chongqing, Beijing and Guangzhou, ranking Delhi at 5th today. This trend of fast-growing cities is widely noticeable in Africa as well, with Lagos in Nigeria being among the fastest growing, and other cities like Kinshasa in Democratic Republic of Congo and Dar Es Salaam in Tanzania.

Li, et al. (2019) analyses **four major problems of Megacities** from urban geography and ecology perspective, as follows;

- ***Land subsidence*** due to over exploitation of groundwater.
- ***Environmental problems*** such as, pollution, urban heat islands, urban air quality and haze, carbon emissions and dust storms, etc.
- ***Traffic congestions***, parking difficulty, public transport. This is closely related to the above problems of pollution, haze, etc.
- ***Energy consumption and production*** that is inefficient and unsustainable.

However, there is also the problem of declining of urban population. While large influx of population into urban boundaries is on rise, most of these cities also face high risk of population decline and loss of life due to geographical location, mostly coastlines, and consumption patterns are the vulnerable to at least one of the 6 types of natural disasters – cyclones, floods, droughts, earthquakes, landslides and volcanic eruptions. Some of the factors that contribute to the high risks of megacities in developing nations are; high population exposure due to housing concentration, complex and ageing infrastructure, lack of robustness of critical facilities and weakness of preparedness for response and relief (Wenzel et al., 2007). In addition, stagnating population has been associated with low fertility rates, particularly in Europe.

The mass movement of population from rural to urban has led to drastic changes in the constructed habitats and has put ever increasing demands and stress on infrastructure, limited resources and inequitable access, and deteriorating living conditions, leading to a large chunk of the population to be below poverty line and without social welfare. While cities today approximately occupy only 2% of the total land, it contributes towards 70% of the GDP and is responsible for 60% global energy consumption, 70% greenhouse gas emissions and 70% global waster generation. The enormous requirements of natural and human resources for energy, food, industrial production, services, construction, infrastructure, and maintenance, causes severe ecological impacts. In turn, development sprawl encroaches ecologically sensitive areas, such as, estuaries, coasts, riverbeds, which in turn require high developmental costs and maintenance more expensive. Other anthropogenic hazards, such as terrorism and war are also largely targeted towards megacities.

Fig. 3.1 : Megacities of the World (source : United Nations)

Krass (2007) noted **how megacities affect global changes**, enlisted below, and in turn, are affected by them. These changes maybe,

- ***Geo-ecological change:*** pollution, sea level rising, global warming, etc.
- ***Geo-economic change:*** globalisation, international labour division, urban markets, etc.
- ***Geo-social change:*** urban disease, social justice, migration, human security, etc.
- ***Geo-cultural change:*** urban ethnicity, social movements, global media, urban cultural diversity and hybridity; and
- ***Geo-political change:*** resource security, social stability, social justice, welfare, etc.

To combat these challenges, the '*New Urban Agenda'* was adopted during the UN Conference on Housing and Sustainable Urban Development (Habitat III) in 2016, that outlined the implementation of; Urban rules and regulation, Urban planning and design, and Municipal finance, with consideration of national urban policies to support sustainable development. In addition, megacities are greatly leaning on digital Information and Communication Technologies (ICT), to envision the ever-growing city through 3D modelling and visualisations, image and data fusion, Big Data, IoT and real-time Earth observations, becoming the backbone for 'digital cities of the future' or '**Smart cities**'. However, while megacities are heavily into employing 'Intelligent' strategies, but it may be argued that without addressing the elements of sustainability and noted improvement of quality of life, and only through employment of ICTs is inadequate. Presently, western megacities such as London, Paris and New York, are focusing on infrastructure for quality of life by ensuring; sustainable utilisation of land resources, improvement of design quality of buildings and urban spaces, balance between economic growth and environmental protection, improve air and water quality, climate change, etc. This gives an insight into what should the priority of developing nations such as India be to achieve SDG 11 – Sustainable Cities and Communities.

3.1.2 Smart Cities

SDG 11 aims for inclusive, safe, resilient, sustainable cities and 'smart city' planning is a proponent of the same, as it strives develop frameworks to technologically support all basic infrastructure and services required for its inhabitants towards becoming self-sufficient. Megacities and upcoming cities, with growing issues of sustainability are struggling to preserve natural and economic resources, lean on 'smart city' concepts however, the ***ground implementation is heavily dependent on the level of development, willingness to change and availability of resources*** (Bordoloi and Acharya, 2023).

'Smart' city is a broad concept with various sub-themes; urban and regional planning, economic development, environment and sustainability, ICT and technology. Integration of digital technology with improving urban areas and public spaces, reducing environmental impact, involvement of citizens in policymaking, and utilizing entrepreneurship and human capital for urban development; thereby making networks and services more efficient, flexible, and sustainable for the benefit of its residents is the key characteristic of a Smart city.

The **dimensions of Smart City** are smart - people, smart living, smart government, smart transportation, smart environment and smart economy. These are further expanded to include smart technology, smart infrastructure, smart water and waste, smart agriculture and smart security. In summary, the term 'smart' has three conceptual elements: Technology, including hardware and software infrastructures; People and their associated attributes of creativity, diversity and education; and Community, referring to institutions, governance and policy. And the four city technological brands included under 'Smart City' are 'Digital City', 'Intelligent City', 'Ubiquitous City' and 'Information City'. However, these must not be at the expense of social and environmental impact as 'quality of life' is the eventual goal of smart cities. Therefore, a term **'smart sustainability'** is often used to stress on the need to respect local or planetary limitations and to utilise resources without compromising the needs of future generation.

Thus, Smart City strategies help to create sustainable cities and communities by addressing social problems, optimising financial resources, and mitigating environmental consequences through conservation and controlled use of natural resources. **Key features of 'sustainable smart cities'** include *compactness, population density, sustainable transport, mixed land use, green areas, passive solar design and diversity*. However, there are still gaps between real world problems and available strategies and solutions.

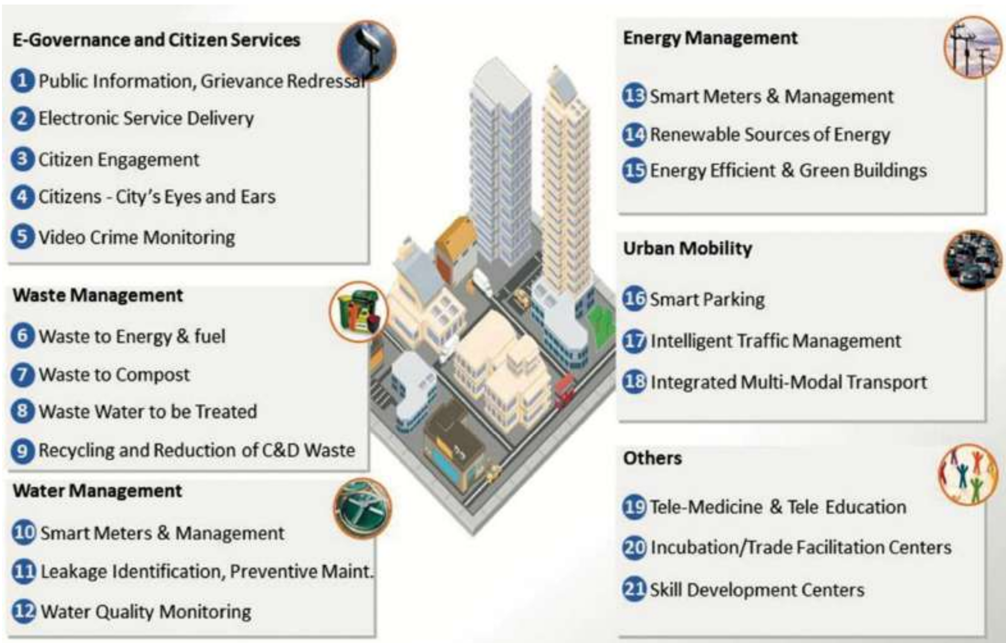

Fig. 3.2 : Illustrative List of Smart Solution, Mission Statement & Guidelines, MoUD, GoI (Jun 2015)

In June 2015, the Govt. of India launched the '***Smart Cities Mission'*** to develop sustainable and inclusive cities, keeping in mind that by 2030, 40% of the Indian population will be in urban areas and will contribute towards 75% of the GDP. The Ministry of Housing and Urban Affairs (MoHUA) outlined the strategic components of a smart cities as; adequate water supply, assured electricity supply, sanitation and solid waste management, efficient public transportation and urban mobility, affordable housing, robust IT and digitalisation, e-governance and citizen participation, sustainable environment, safety and security of citizens, and health and education. The mission envisioned 100 cities across the country and strategized based on area-based development through retrofitting, redevelopment, greenfield development, and pan-city initiatives. There are several schemes, such as, Atal Mission for Rejuvenation and Urban Transformation (AMRUT), Swachch Bharat Mission, National Heritage city Development and Augmentation Yojana (HRIDAY), Digital India, Skill Development, Housing for All, etc., that offer complementarity to the Mission and convergence of these is promoted. Three tiers of monitoring, at national, state and city level is proposed, and while it is presently centrally funded, it requires state government and urban local bodies to contribute equal amounts for implementing the Smart City, approximately 100crore per city per year for 5 years. The total allocated investment stood at Rs.205,018 crore ($ 27.6 Billion) as of March 2021 and a recent Rs.16,000 crore has been allocated in Budget 2023. Presently, MoHUA reported that more than two-thirds of the total 7,804 projects have been completed.

3.1.3 Future vision of Habitats

Future habitats, while predominantly urbane and highly digitalised, will reflect the paradigm of Industry 5.0, with a shift from economic value to social value. This is exemplified by automation, robots and smart machines working alongside humans with resilience and sustainability as priority. The cities of the future, while 'intelligent', will lean strongly on tenets of Sustainability as the backbone for development and aim to create value through harmonising **five types of sustainable capital** from where goods and services are derived, namely, Natural Capital, Human Capital, Social Capital, Manufactured Capital and Financial Capital.

A large number of **futuristic city developments** underway all over the world are;

Amravati, capital city of Andhra Pradesh, India, designed by Foster + Partners, is under the Smart City Mission, is aimed to cover 217 sq.km. such that over 60% of the core area is occupied by greenery or water bodies.

Chengdu Future City, China, designed by OMA, will occupy 4.6sq.km. with focus on smart mobility network and a car-free masterplan.

New Administrative Capital in Cairo, Egypt, designed by SOM, will cover 700 sq.km. and will feature one of the world's largest urban parks.

Smart Forest City, Mexico, designed by Stefano Boeri Architectti, is intended to be a forested smart city near Cancun with plant covered homes and 7.5 million carbon absorbing plants and trees across its 557 hectares (5.57 sq. km).

Fig. 3.3: Future Projects - (*Top Left*) *Amaravati,* India, by Foster+Partners, (*Top Right*) *Chengdu Future City*, China, by OMA, (Bottom Left) *Maldives Floating City*, Maldives, by Waterstudio, (*Bottom Right*) *The Line*, KSA, by Neom

Several attempts to use advanced building and construction technology is also underway to **reclaim or rebuild presently vulnerable cities**, such as;

BiodiverCity, Malaysia, also designed by BIG, is a 1821 hectare (18.21 sq.km) development of three artificial islands built of the shore of Penang Islands, built using combination of bamboo, timber and concrete.

Maldives Floating City, Maldives, designed by Waterstudio, as a response to the rising sea level and the concern of the Maldives becoming uninhabitable by 2050, will be built on hexagonal structures that rise with the sea.

A prototype climate-resilient floating city is also being designed by BIG and tech company***, Oceanix, off the sea of Busan, South Korea***.

Further, state of the art, **breakthrough proposals** are also underway in unoccupied land stretches with the intention to design for the future population and a sustainable way of living, such as;

The Line, Saudi Arabia, which will house a 170km long, 500m tall, but 200m wide, linear city stretching to touch the sea, and is envisioned as a model for ‘nature preservation and enhanced liveability’.

Telosa city, USA, designed by Bjarke Ingels and BIG on an unoccupied 150,000-acre (607.03 sq.km) site in the western desert.

Design of green buildings, use of eco building materials, incorporation of natural environment and urban open, green spaces, apart from the 'smart' services and their allied infrastructure would be characteristics of cities of the future. In addition, there is a hope that the citizens of the future will be more conscientious as a society and continue to strive towards the SDGs collectively. This brings us back full circle to the early '**social indicator movement'** of the 1960s where the important linkages between infrastructure and quality of life (Terleckyj, 1975) was captured in order to systematically assess existing possibilities for social change, by mapping the infrastructure against attributes of human habitat, i.e., Health, Safety, Recreation and Aesthetics, Economic opportunity and Leisure. Aschauer (1989) reflects that, *the general public desires cleaner environment, safer urban streets, increased mobility and economic opportunity for the disadvantaged, and an economy well equipped to compete internationally*; and empirically establishes the importance of infrastructure development and investments towards improved quality of life and economic performance.

3.2 TRANSPORTATION

The global economy functions through trade and commerce, and transportation service sector is the most essential aid. It not only supports the supply chain through timely movement of raw materials and resources, but also enables eventual distribution of finished goods, in turn assuring economic competitiveness and growth. It is also the mode of travel for people for employment and education opportunities, healthcare, leisure, or social networking, etc. But while this was greatly hampered in the pandemic era, with millions with home bound. The times are also exemplar of the heights of exploitation of transportation infrastructure and logistics, becoming one of the most critical infrastructure sectors along with Energy, Water and Telecommunication. And today's new normal is quick and easy, door-to-door delivery, demanding development of further land, water, underground, air, as well as futuristic modes of transport infrastructure.

3.2.1 Land Transport Infrastructure

Land transport can be broadly categorised into roads, railways and metros; a 'way' for travel. The word 'way' as per Britannica, stems from the Latin '*veho'* which means, "*I carry"*, derived from the *Sanskrit* word, '***vah'*** meaning to "*carry, go or move*".

Roads

The generic word 'road' used to encompass all land vehicular ways is derived from the Old English word '*rad'*, which means "*to ride*". However, the term highway predates it, back to the elevated Romans roads which created a mound in the centre and ditches on the side, upon compacted layers of soil and gravel. It is used to refer to major roadways that connect several rural and urban space and is characterised by various controlled points of entry and exits for traffic. While 'streets' has its origin in Latin '*strata'* meaning paved, and is till date used to refer to all small rural scale roads.

Road networks follow a system of hierarchy in their capacity and capabilities. Most roads world over are conventional, undivided two-way; however, there are divided roads such as; expressways, having minor at-grade intersections; freeways, having no at-grade intersections, collectively called motorways in the UK. Functionally roads are classified as: Highways - the major roads between regions; Arterial roads – that carry through traffic from adjacent areas and are major roads within a region; Collector, distributor and feeder roads – that carry only through traffic from a certain area perimeter; and local streets – that do not carry through traffic but only serve adjacent properties.

The earliest records of animal trodden paths used by man have been dated back to 6000 BCE, with evidence on first constructed road dating 4000BCE at Ur, present day Iran. Oldest existing paved road, made of layers of sandstone bound by clay-gypsum mortar with two rows of basalt slabs in the centre for use by foot, while the dipping edges were for animals, was built by the Minoans on the island of Crete. It was about 50km long at an elevation of 1300metres, from Gortyna to Knossos. The earliest long-distance road, traversing 2500 kms, between Susa in the Persian Gulf to the Mediterranean ports of Smyrna (Izmir) and Ephesus, was first used around 3500 BCE but came into organised use aroud 1200 BCE under the Assyrians. The Egyptians around 2600 BCE built roads to haul limestone blocks for pyramids and to connect towns and temples. Routes from Thebes and Coptos on central Nile ran towards the Red Sea and from Memphis (Cairo) towards Asia Minor. By 1500 BCE, a trading network across eastern and central Europe called **the 'amber routes'** grew for trading various materials such as, amber from northern Europe, salt from Austria, freestone from Belgium, lead and tin from England, flint from Denmark,etc. and connected Hamburg, Cologne, Frankfurt to Passau and Venice. In 334 BCE, Romans championed road making building nearly 85,295 kms, radially connecting Rome to its several provinces, with 29 great military roads (*viae militares*). **The Appian Way** was the most famous one, constructed in 312 BCE and at a distance of almost 660km from the capital, along the Mediterranean coast. In addition, the Romans also had two classes of public transport – the express service and freight service. China too had an extensive road system by the 7th century CE across 40,000km, development of which began under the Emperor Shinhuangdiin 220BCE. In India, under the Mauryan Empire around the 4th century BCE, had quality roads across its empire which stretched from Indus River in the Northwest to the Bramhaputra in the East, from Himalayas in the North to the Vindhyas in the South. Their **Great Royal Road** ran through Taxila or Takshashila, in modern day Afghanistan, across Punjab and Prayagrai, until the mouth of the Ganga River. The famed **'Silk route', the longest road of the time** that linked the Mediterranean with China, is said to have existed partially around 300 BCE and by 100 BCE it was pivotal to trade. However, the following centuries saw neglect of road systems without strong dynasties and leaderships, and wasn't until 8th century that road revival became a priority. In 9th century that the Moors of Cordoba, Spain developed an extensive street network. The following 11th and 12th century witnessed revival of cities and roads, mostly in western Europe, and by 15th century, well maintained infrastructure was ubiquitous. Meanwhile, in South America, the Inca empire (1000 – 1450 CE) extended the '*Qhapaq Nan'* or royal road from Ecuador, Peru until Chile, with two parallel systems – one along the coast and another along the Andes. Eventually, by 18th century, the modern masters of road building – Treaguest, Telford, Mc Adam, Mitchell, appeared in England and France.

Road design is an exceedingly important aspect of national and regional planning, especially in the context of urban connectivity and the population, commerce, industry and transportation needs of the community. Estimating traffic on a route as well as conducting civil surveys to establish the site conditions are integral to successful road system planning and design. To design a road, three-dimensional road alignment of the cross-sectional profiles along the terrain on which it is laid is of utmost importance. For quality and uniformity, **standards and codes** are established for different types of roads. It helps with guidelines on, number of traffic lanes as per traffic volume and speed, width of lanes, carriageways and shoulders, and specifications on roadside barriers. Other integral elements of road system design include design of pavements and drainage. Safety and serviceability of the roads are also critical aspects to be considered and financing and maintenance are key determinants. Jurisdiction of the section of the road and its onus, also plays an important role in case of any legality and taking responsibility. In India, roads are under the aegis of Ministry of Road Transport and Highways. Some of the **salient features reported** in the review 2022, note the major push in the following areas;

- Highway development
- Connectivity
- Logistics development
- Online citizens service
- Under Bharat series, regularisation of registration mark
- Retro fitment of CNG and LPG kits allowed in BS VI
- Bharat N-CAP safety rating introduced
- Compensation of victims increased
- Green fuel and vehicles
- Construction of '*amrit sarovar*' along National Highway
- NHAI InvIT bonds enlisted on BSE and NSE, and
- Manthan conference held in Bengaluru to discuss issues in road, transport and logistics.

Railways and Metros

In earlier chapters, we have discussed the oncoming of railways. The role civil engineer is pivotal for railway transport infrastructure, and encompasses surveying for a new line, construction and maintenance of the line, ensuring longevity, safety and reliability of the structure. In addition, the civil engineer also has to design bridges over rivers, station buildings and allied facilities, such as, office rooms, parcel offices, goods sheds, restrooms, wating lounges, as well as, loco sheds, pump houses, water and drainage lines, etc.

Railways evolved from **'tramways'** which were originally of stone slabs, timber and baulks, laid in flush with the road surface for horse carriages, and later, reinforced with iron straps or plates. Further on, these were improved, and tracks were designed, having angle irons with a vertical leg, later replaced by cast iron beams, which further evolved into rail sections on which the locomotive's wheels align. The wheels transfer the load of the locomotive on to the two rails of the track, kept at specific distance, i.e., gauge, and is placed on perpendicular sleepers equitably distanced on a bed of ballast. There are ballast-less, and continuous longitudinally-supported tracks as well, the latter being very uneconomical. Earlier, rails were of 'I' or dumb-bell sections; but this further developed into a 'T'-section, laid inverted and popularly termed as the 'Flat-footed' rails. Other rail profiles are Bullhead rail, Grooved rail, Bridge rail (inverted U-shape) and Barlow (inverted V-shape) rails. There are two common types of gauge - Broad gauge (1676 mm) that supports the speed of 100-160 Km/h, and Metre gauge (1000 mm) having a maximum permitted speed of 75Km/h. Another type is Narrow gauge, such as those of the Darjeeling Himalayan Railway or the Toy train which has achieved the UNESCO World Heritage status. Sleepers maybe made of wood, cast iron, steel, concrete. The rails are fastened on to the sleepers with dog spikes, and the ends are connected by fish plates or fish bolts, and can be switched to direct the trains. Other types of fastening are - round spikes and screw spikes, different types of bearing plates, tie bars, etc. The rail line is either placed on an embankment, or in a cutting where a pit is dug and the centre is raised to house the track, as it cannot be laid directly on ground. The railway track comprises of rails, sleepers, fastenings and ballast, and may also be called **'permanent way'**.

The **first steam locomotive** to carry passengers, designed by engineer, George Stephenson, began operations between Stockton and Darlington in 1825. The first railway proposal in India under the British rule was made in 1832, and the **first train transport named 'Red hill'** plied in 1987 carrying freight of granite for road -building. India's inaugural passenger train operated by the Great Indian Peninsular Railway, having three steam locomotives named '*Sahib, Sindh and Sultan',* ran between Bori Bunder (Bombay) and Thane, for a distance of 34kms, in 1853, and first passenger train ran between Howrah and Hoogly in West Bengal in 1854. Today, the Indian Railways is Asia's largest network and among the world's largest. It is about 108,706 track Kms and runs around 11000 trains daily, off which 7000 are passenger trains carrying around 13 million passengers every day.

In 1984, South Asia's first subway line began operations in Kolkata after 23 years since the commencement of underground construction. Delhi, too, had a urban rapid transit proposal back in 1969 but the construction began in October 1998. Today, the Delhi Metro is a benchmark, not only as the largest and busiest metro system, but also due to its state-of-the-art design and construction. The network consists of 10 colour coded lines, covering the National Capital region and its satellite cities of Ghaziabad, Faridabad, Noida and Gurgaon. It is a mix of broad and standard-gauge and has underground, at-grade (at road level) and elevated sections. **DMRC (Delhi Metro Rail Corporation)**, a company with equity from Government of India and Government of Delhi, under leadership of E.Sreedharan, has been certified by U.N. as the first rail-based system in the world to get carbon credits for reducing greenhouse gas emissions and carbon emissions. Mumbai also boasts of the first of its kind, Mumbai Monorail, that was completed in 2014 and is presently witnessing the development of the sixth longest operational metro network in India.

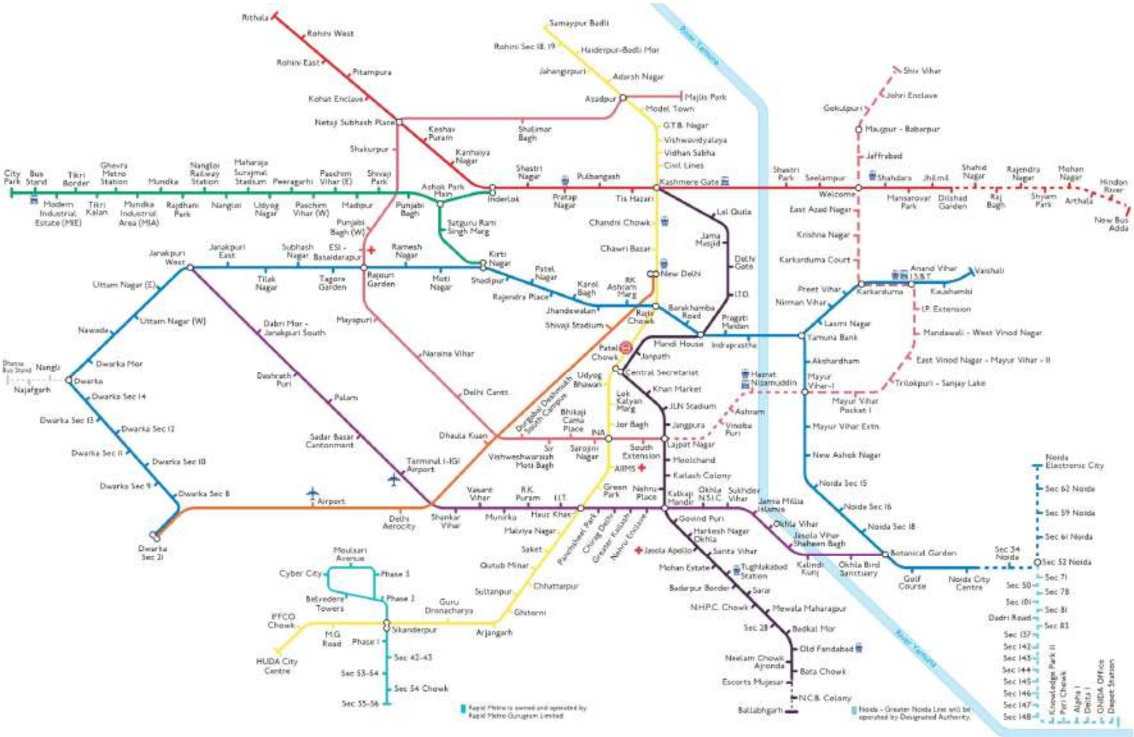

Fig. 3.4 : Delhi Metro Map (source : India Tourism, 2022-23)

Tunnels : Below ground or under water

Tunnels are under ground or under water, enclosed passages constructed by digging through rock, soil and earth, for movement of vehicles. With the massive excavation needed, the integrity of the passage is subject to the loading and pressure of the surrounding earth and/or water, construction of tunnels is highly specialised. There are various methods of tunnel construction on land, such as, *Cut and cover method* where a trench is cut and covered by some support;, *Bored method* where boring machines are used; *Clay kicking* method common from the time of World War I, where clayey soil is literally kicked to create a tunnel; *Shaft method* is used for reaching great depths from the ground level and pre-cast shaft structures with concrete walls are lined and connected; *Pipe jacking method* where pipes are driven underground using hydraulic jacks, usually under existing infrastructures like roads, railways, etc.; and *Box jacking method* where boxes instead of pipes are driven in. For underwater tunnel construction usually calls for immersing tubes or boring tunnels through rock.

The **world's longest road tunnel** is the *Lærdalstunnelen*, of 24.5 Km linking Aurland and Lærdal, and offering a ferry free connection built in 2000, followed by the Yamate tunnel in Tokyo, Japan measuring 18.2Km built in 2015 and the Zhongnanshan in Shaanxi, China measuring 18.04Km built in 2017. The Eisenhower Road tunnel in Colarado is one of the **world's highest tunnels** at a height of 3,401m above sea level, on the Rocky Mountains.

The **world's longest and deepest train tunnel** is the Gotthard Base Tunnel in Switzerland, running 57 Km in length and at a depth of 2300m under the Swiss Alps, significantly reduced train travel time connecting Zurich to Milan. Previously this distinction was held by the Seikan railway tunnel, spanning the Tsugaru strait connecting Honshu and Hokkaido, running 53 Kms at 140m depth. Another phenomenal feat is the Channel tunnel across the English Channel between England and France, spanning 50Km, having two rail and one service tunnel for passenger and freight.

A piece of modern engineering is the **SMART** (Stormwater Management and Road Tunnel) Tunnel in Kuala Lumpur, Malaysia, designed keeping in mind the flash flood situation that the city faces and operates in three ways : as a road tunnel when there is no flood, the upper level operates for traffic while the lower channel allows water diversion when there are medium floods; and closes for traffic but allows floodwater to flow through via a holding pond, bypass tunnels and a storage reservoir during heavy floods.

3.2.2 Water Transport Infrastructure

Seaports

Ancient Greeks heavily relied on sea travel as land travel was difficult. Greece is a series of archipelagos and peninsulas, surrounded by the Ionian, Mediterranean, and Aegean Sea, which they traversed by trireme ships. Beyond travel, they instituted maritime trade and frequented the ports of Canopus before Alexandria in Egypt and Messina in Sicily from Athens. Ostia Antica and Portus were later set up in Rome, and Swahili kingdoms of East Africa were described to have vibrant trade ports. Trade flourished amidst the Arabs, Greeks, Romans, Egyptians, Africans and Chinese. The south-western coastal ports of Muziris and Calicut, or Kozhikode, in present day Kerala, are accounted in ancient texts and is to have played a crucial role in the spice trade. The latter gained prominence after the arrival of Vasco da Gama. The port of ***Lothal***, the southernmost city of the Indus civilisation, at the edge of the Arabian Sea in present day Gujrat and is believed to be **the world's oldest dock**, dating back to 2200 BCE. Another important port in Gujrat, at the mouth of the river Narmada is Bharuch. East of the Indian subcontinent, Chittagong in present day Bangladesh, has been found referred in Ptolemy's map, dating back to 2nd century. And the south-eastern ports of Tuticorin (Thoothukudi), Arikemedu at modern day Pondicherry and Poompuhar, all in Tamil Nadu were bustling ports. In the far east, ancient seaports were Guangzhou in China during the Qin Dynasty and Osaka in Japan during the Edo period.

Civil engineering plays a crucial role in the design of water transportation and construction of ports and harbours. Ports are locations where ships and vessels can dock and allow movement of people and goods and are either on a coasts or shores, while harbours are constructed for the safe keeping of the vessels. Other infrastructures that have nuanced variations but support sea travel are; quay or wharf, pier, jetty , berth and dock, and all together form a network of waterway infrastructure for overseas transportation.

Inland waterways and canals

The beginning of rails marked the demise of canals and inland waterways, which used to be the preferred mode of transportation. All ancient civilisations settled around rivers, such as, the Indus civilisation along Indus River, Mesopotamia between the Tigris-Euphrates, Memphis (Egypt) along the Nile. Several prominent European capital cities, such as London, Paris and Amsterdam all are also along rivers. Apart from being a source of water for consumption and irrigation, riverways and inland waterways became the mode of trade and travel. Several other cities, such as, Giethoorn in Netherlands, Birmingham in England, Burges in Belgium, Hamburg in Germany, Stockholm in Sweden, are all traversed with network of canals with most prominent being **Venice**, Italy, which has **150 canals** including the *Grand Canal*. These were built by lining the dugout pits by closely spaced alder wood which is waterproof, to make the lagoon fit for habitation. While most of these canals are 1.5-2m deep, the *Canale della Giudecca* that separates the main part of Venice from the island of Giudecca, is about 12-17m deep.

Two of the most ambitious canal projects that are engineering feats are; the **Suez canal**, also referred to as *Qanat al Suway* (length of 193Km with branches, depth of 20m and width of 205m) across the Isthumus of Suez in Egypt which connects the Guld of Suez and the Mediterranean, thereby allowing a quicker path between the Pacific and Indian Oceans, and the **Panama canal** (length of 82Km with branches, depth of 12m and width of 150m) which connects the Pacific and the Atlantic Oceans across the Isthumus of Panama amidst the Caribbean. These two interventions significantly changed the time required and ease of transport for trade, becoming pivotal contributors to economic growth.

The **world's oldest and longest man-made canal** is the '*Great Canal'* or the Beijing-Hangzhou canal that connects the Yangtze and Yellow He rivers, across 1,782 – 2,470 Km with branches in length and varying width between 40-350m, having a depth of 2-3m. It is adorned with 21 gates and 60 bridges and has been recognised by UNESCO In 2014.

3.2.3 Air Transport Infrastructure

Aeroplanes, helicopters, light aircrafts, hot air balloons, blimps and gliders, drones or UASs (unmanned aircraft systems) are all vehicles of air transport and require certain physical infrastructure to support flight and while not in flight, such as service, maintenance and parking.

Airports are complex transportation hubs and is spatially divided into three: airside, landside and terminal that connects the two. The civil engineer has the responsibility of structurally designing and constructing; the airside layout comprising of runways, taxiways, parking aprons, lighting and signages, navigational and visual aids; the landside facilities, such as parking lots, fuel tank farms, access roads, technical buildings like control towers for ground aid, etc. and passenger and cargo terminals.

Helicopters too have designated bases with fixed operations and services like customs, fuel bunkering and maintenance, called heliports. It is like a small airport specifically for helicopters and other vertical-lift vehicles. Often high-rise buildings, hospitals and other buildings or campus of importance have helipads for landing and take-off only. This infrastructure is of particular importance as helicopters and drones are the preferred mode of transport during natural disasters for rescues and searches, supply as well as surveillance.

Aviation falls under critical infrastructure in India and has been a hot target for attacks, online and offline. The regulatory body for India is the Directorate General Civil Aviation (DGCA), empowered by the **Aircraft Act 1934**, implements standards and recommended practices of the International Civil Aviation Organisation (ICAO), and later further bolstered by the Aircraft Rules 1937, is authorised to specify requirements and compliance procedures through Civil Air Regulations (CAR). It outlines the operations and planning of infrastructure as well. The DGCA also regulates the airspace and in turn, monitors and supports the use and manufacture of UAS, as well as authorises remote pilot training and certifications, to overall ensure National security.

3.2.4 Futuristic Transport Infrastructure systems

'Smart' transportation is the vision of the future, with driverless cars, flying taxis, delivery drones and levitating trains being already on the horizon. **Novel concepts** of futuristic transportation, impose the need for reimagining the infrastructure, such as;

Self-drive Cars or autonomous vehicles (AV), a technology that is seemingly possible in the near future, require physical and technological framework to support and enable the operation of autonomous vehicles. The most important element is the need for communication network with real-time data exchange through vehicle-to-infrastructure (V2I) systems, such as sensors in roads or street signs that send signals to AVs, helping them navigate city streets. There are also alternatives like, dedicated short-range communications (DSRC) and cellular vehicle-to-everything (C-V2X) systems. Physical infrastructure is also crucial, such as, parking facilities equipped with autonomous parking capabilities, maintenance stations specifically designed for self-driving cars, and a comprehensive network of charging stations as many self-driving cars operate on electric power, to support uninterrupted operation.

Maglev, a technology where magnetic levitation actuated by two sets of opposing electromagnets along the tracks, allows trains to travel at high speeds. Shanghai Transrapid is the presently the world's fastest commercial electric train and goes up to 431 Km/h. Japan began development and construction of SCMaglev under the Central Japanese Railway Company in 1969, and the HSST (Linimo) in 1974. A new maglev line, named the Chuo Shinkansen started in 2014. In 2016, South Korea inaugurated the Incheon Airport Maglev. Currently, all maglev trains are in use in Asia.

Hyperloop, "*an ultra-high-speed transportation system in which passengers travel in autonomous electric pods*" futuristic transportation concept. It was first conceptualised by Elon Musk in a white paper in 2013 for intra-city mass transit and envisioned "*a tube, over or under the ground, that contains a special environment*". It consists of low-pressure tube with capsules that are transported at both low and high speeds throughout the length of the tube, such that the capsules are supported on a cushion of air and are accelerated by a magnetic linear accelerator affixed at various stations.

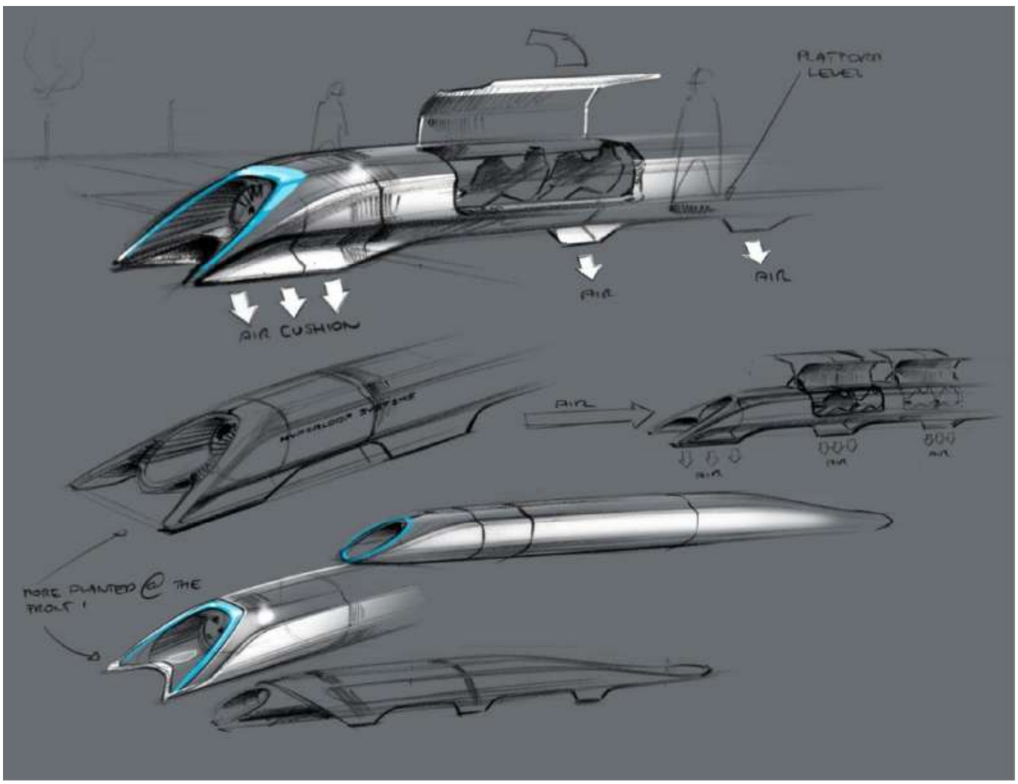

Fig. 3.5 : Hyperloop Alpha concept sketch (source: www.tesla.com)

Underground tunnel roads are another vision of Musk for futuristic transportation where he conceptualises a network of underground tunnels equipped with trolley like platforms to lower the cars from the surface and speedily transport them to their destinations, alleviating congestion. This is much in line with the idea of autonomous vehicles, driverless cars, and flying taxis, which are all well underway in prototyping phase.

The motivations of these proposed futuristics concepts are; reduction of congestion and greenhouse gas emission, reduce fatalities and prioritise safety, provide access to disadvantaged and to improve on time. 'Quality of life' and low environmental impact are key and in turn, requires to re-imagine the supporting infrastructure for energy, telecom, etc.

3.3 ENERGY

Global energy infrastructure includes delivery systems of oil and natural gas pipelines, power generation facilities and transmission lines, storage facilities, etc. Inclusive economic growth, clean air, climate has necessitated innovations in efficiency gains, new production flows, digitalization, smart grids, etc., requiring infrastructure to become sustainable, resilient, and secure (International Energy Forum). Geopolitical and natural events and accidents may lead to disruption of energy markets, and hence, policy cohesion and regional market integration through cross-border networks are essential to meet SDG, i.e., Ensuring access to clean and affordable energy.

"*Energy generation depends on a country's natural resource endowments and technology to harness them*" as per Government of India Budget report 2012-13. India is the world's third largest producer, as well as, consumer of electricity, but this is predicted to increase by 4.5% annually to 2035. Presently, India's energy consumption is 1010kWh per capita against a world average of 3200kWh. Gujrat, Maharashtra, Tamil Nadu, Rajasthan, Karnataka, and Uttar Pradesh are the top power producing states of the country. However, the key issues are poor transmission and distribution grids. India has both non-renewable reserves, of coal, lignite, petroleum, and natural gas, as well as renewable energy resources of hydro, wind solar, biomass, and cogeneration bagasse. But while the non-renewable resources are poor, India is richly endowed with renewable resources, and several policies and initiatives are presently in place to leverage that.

3.3.1 Renewable Sources

Renewable energy, popularly referred to as **'clean' energy**, comes from natural sources or processes that have a higher rate of replenishment than its consumption. Their sources are sunlight, wind, water, geothermal and biomass. While large hydroelectricity projects and biomass levy a trade-off on wildlife, biodiversity and climate change, other sources have very little negative impact on the environment and in turn, generate lower emissions.

Solar technologies can also provide heating and cooling solutions, natural lighting, and fuel for cooking. Solar power can be harnessed through mirrors and photovoltaic (PV) panels, that concentrate the solar radiation and converts it into usable fuel or electricity. PV cells are made of silicon and present technology has not only significantly reduced its prices, but extended its life to up to 30 years. Solar farms are fast becoming a large initiative where PV cells are splayed, often aided by mirrors to concentrate the sun light, and in some cases, wastelands, water bodies and wastewater facilities are employed.

Wind energy exploits the kinetic energy of moving air by using large wind turbines situated at on-shore or off-shore locations, as per the average wind speed of the location.

Geothermal energy is garnered through the heat extracted from the earth's interior reservoirs using wells and ducts. Hot fluid of varying temperature is pumped through turbines to generate electricity, and recycling of the used water and steam ensures low emissions.

Hydropower harnesses the energy of falling water from higher to lower elevations and can be generated from reservoirs or flowing rivers. Beyond providing electricity, these sources also provide water for irrigation and drinking, as well as help in flood and drought control, and navigation services. However, rate of precipitation and annual rainfall impacts the functioning of hydropower.

Tidal and wave energy of the ocean is also a potential source of clean energy and is generated through tidal barrages which are dam-like structures located in ocean bays and lagoons, or via devices which are ocean floor-anchored or placed just below the wave surfaces. These interventions may cause adverse effects on the life under water.

Bioenergy is produced from biomass which is the organic remains of various sources, such as, wood, charcoal, dung, manures, reside from agriculture and forestry, and other organic wastes. It is used mostly in rural areas for cooking, lighting, heating. However, in spite of its natural origin, it has detrimental environmental impact as it produces emissions and bioenergy plantations may lead to deforestation and land-use change, making it arguably not a 'clean' energy source.

India is the **third largest producer of renewable energy** and is rich in clean energy sources, such as, solar, wind, and small hydro, with high potential for energy generation. India globally ranked 4th in solar power capacity, wind power capacity and overall renewable power installed capacity as of 2021 and ranked 6th for hydropower generation in 2019. Today, as of January 2023, India's installed renewable energy capacity, is at 40.9% of the total installed power capacity. Solar (63.3 GW), hydropower (46.85 GW) and wind (41.9 GW) are the largest contributors followed by biomass (10.2 GW), small hydro (4.92 GW) and 'waste to energy' (0.52 GW) as contributors towards the renewable energy capacity.

The past year has been pivotal for development of renewable energy. The union cabinet approved 2,614 Crore investment in the 382 MW Sunni Dam Hydroproject by SJVN (Satluj Jal

Vidyut Nigam). SVJN also signed an agreement with the Utter Pradesh government to implement 3 solar power project and commissioned a Solar power project (75MW) at Parasan Solar Park near Kanpur, U.P. They further signed up with Tata Power Solar systems to build a 1000MW solar project near Bikaner and announced collaboration with Govt. of Assam for development of hydro and renewable energy projects. National Hydroelectric Power Corporation (NHPC) and Govt. of Himachal Pradesh intend to implement a 500 MW hydroelectricity project in Chamba District, H.P., as well as two project in Nepal. Tata Power Green Energy Limited (TPGEL) commissioned a hybrid power project in Rajasthan and was awarded a solar project in Solapur, Maharashtra. National Thermal Power Corporation (NTPC) announced partial power generation from the floating solar energy plant in Kayamkulam, Kerala, and Jetsar, Rajasthan. Adani Solar and Smart Power India (SPI) signed an MoU to promote usage of solar rooftop panels in rural India. To further promote this the Ministry of New and Renewable Energy has developed a national portal for citizens to directly apply. Several **policies and programs**, such as, '*National programme on High Efficiency Solar PV Modules*', 'smart metre deployment' for National Smart Grid Mission, several electrification schemes and issues of sovereign green bonds and conferring infrastructure status to energy storage systems, are underway to improve India's power sector.

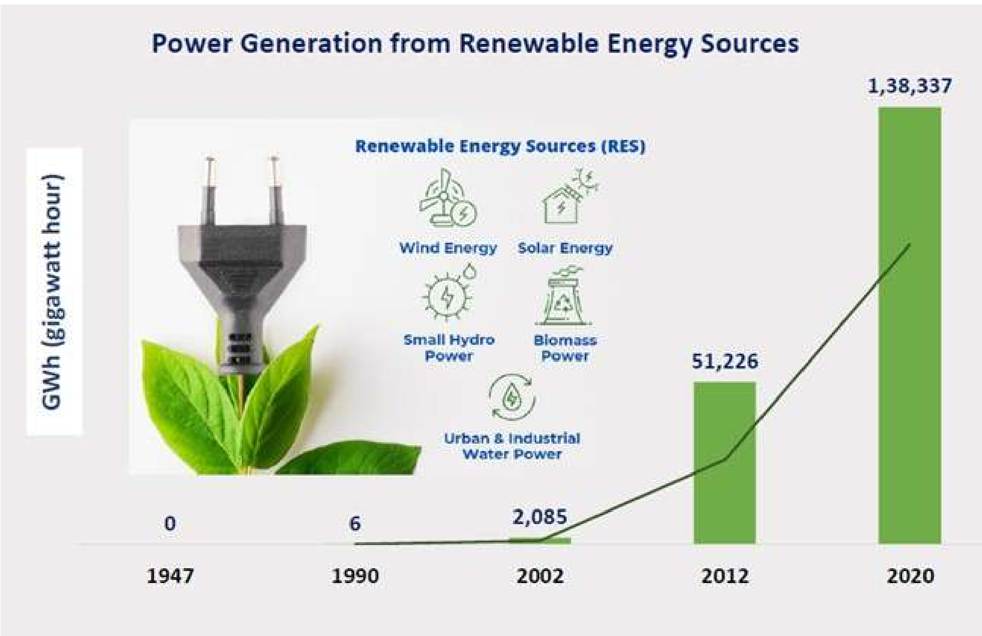

Fig. 3.6 : Power Generation from Renewable Energy Sources (source: Press Information Bureau, GoI)

3.3.2 Non-renewable Sources

Non-renewable energy resources are coal, oil or petroleum, natural gas, and nuclear energy, and are those resources that cannot be replenished easily in consideration of the rate of consumption of the same. Fossil fuels are the largest contributors towards global greenhouse gas and CO_2 emissions, and accounts for over 80% of the world's global energy production. These fossil fuels were formed over millions of years due to the immense heat and pressure within the earth's crust that converted plant and animal matter into coal, oil, and natural gas. The radioactive elements required to generate nuclear energy, usually uranium, comes from mined ore. However, beyond the issue of limited supply, it is the release of high amounts of CO_2 that makes non-renewables an undesired source of energy.

Over 80% of India's requirement is met by coal, oil and biomass, and in spite of having low per capita energy consumption, **India is the 3rd largest global emitter** of CO_2. While the largest domestic source of energy is coal, about 70%, the cumulative domestic fossil fuel production is the lowest among the emerging economies. India relies on crude oil imports as it has only 0.6% and 0.4% of the world's gas and oil reserves respectively. Interestingly however, it is a large net exporter of refined products, amounting to $1.1billion worth to Australia in 2016-17, and has experienced overcapacity in thermal energy. 58.6% of this thermal energy is produced from coal, while the other sources are lignite, diesel, and gas.

3.3.3 New sources and technologies

A promising alternative to traditional fossil fuels is **Hydrogen** as a new source of energy. Hydrogen fuel cells are devices that generate electricity by combining hydrogen and oxygen to produce water and heat as by-products, and can be powered by hydrogen produced from renewable energy sources. They are highly efficient, with an energy conversion rate of up to 60%, compared to around 30% for traditional internal combustion engines. They also produce zero greenhouse gas emissions, which can help to mitigate climate change. However, there are several challenges associated with the adoption of hydrogen fuel cells, such as, the infrastructure required for producing, storing, and distributing hydrogen, and the development of a sustainable and scalable hydrogen production system as the production of hydrogen is currently reliant on fossil fuels. The cost is also a barrier to widespread adoption, although research and development are helping to reduce costs.

Another potent technology for energy generation is **nuclear fusion**, which involves combining of light atomic nuclei to form heavier ones and in the process, releasing vast amounts of potential energy. Unlike nuclear fission, which is currently used in nuclear power plants, nuclear fusion does not produce long-lived radioactive waste and has the potential to provide a virtually unlimited source of clean energy. However, there are several challenges associated with developing nuclear fusion as a practical energy source. Firstly, the high temperature and pressure required to initiate and sustain the fusion reaction are difficult to achieve and maintain for extended periods of time, and the magnetic fields required to contain and control the fusion reaction are complex and difficult to engineer. Secondly, since most nuclear fusion research focuses on using deuterium and tritium - two isotopes of hydrogen, as fuel, the development of a sustainable fuel source is a challenge as tritium is radioactive and must be produced artificially. This requires a supply chain that is currently not fully developed and is also expensive. Furthermore, the development and construction of a commercial-scale fusion reactor would require significant investment and infrastructure.

3.4 WATER RESOURCE MANAGEMENT

Water resource management is a critical to ensure sustainable and equitable access to water resources for human and ecological needs. In recent years, the issue of water scarcity and the need for effective water resource management has become more pressing due to factors such as population growth, climate change, and increasing water demand from various sectors. The World Bank defines Water Resource Management (WRM) as, "*the process of planning, developing, and managing water resources, in terms of both water quantity and quality, across all water uses. It includes the institutions, infrastructure, incentives, and information systems that support and guide water management.*"

The main challenge in water resource management is the need to mitigate the competing demands of various stakeholders, such as, domestic, industrial, agricultural, and environmental. The traditional approach to water resource management has been to focus on supply-side solutions, such as building dams and reservoirs, to increase the availability of water. However, this approach has been criticized for its negative social and environmental impacts, as well as its limited effectiveness in addressing water scarcity in the long term. Recent approaches emphasize the importance of demand-side management strategies, which aim to reduce water consumption and increase efficiency through measures, such as, water conservation, water pricing, and water reuse, which can be effective in reducing water demand, while also providing economic and environmental benefits. Integration of hydrological modelling and remote sensing technologies is also being employed to improve water management decision-making, as these tools provide valuable information on water availability and usage, as well as help to identify areas of water stress and potential risks to water security. However, the role of institutional and governance arrangements in water resource management is pivotal, as effective governance structures and policies are essential for ensuring equitable access to water resources, as well as promoting sustainable and efficient use of water. Participatory and inclusive decision-making processes, as well as the need for greater collaboration and coordination between different stakeholders are much needed to ensure a holistic solution.

3.4.1 Water, Sanitation and Hygiene (WASH)

Ensuring access to water and sanitation is one of the critical SDG (goal 6) as safe water, sanitation and hygiene (WASH) is a basic human need. However, contamination and reclaiming of natural water bodies, overuse of ground water and hindering its replenishment, overall poor management and low investments in water infrastructure are some of the key causes of scarce and unsafe water, and inadequate sanitation.

WASH is a collective term that refers to *a set of interventions aimed at improving access to safe water, adequate sanitation, and proper hygiene practices, and is essential for promoting public health, preventing the spread of disease, and reducing poverty*. **'Water'** stands for access to clean drinking water for household and community use, as well as for agriculture, industry, and other economic activities, **'Sanitation'** implies access to safe and hygienic toilet facilities, as well as the safe disposal of human waste, and **'Hygiene'** refers to practices that promote good health, such as handwashing with soap, safe food handling, and proper menstrual hygiene management. WASH interventions can have significant health and social benefits, particularly for vulnerable populations, especially, women and children. For example, access to safe water and sanitation can reduce the incidence of waterborne diseases, such as, diarrhoea, cholera, and typhoid fever. Improved hygiene practices can also reduce the risk of infectious diseases, improve nutritional outcomes, and promote overall well-being. In turn, it is crucial for achieving several other SDGs , particularly those related to health, gender equality, education and economic growth.

Fig. 3.7 : Systemic support required for successful WASH initiatives (source : www.sdgs.un.org)

WASH implementation in India has been a major focus area for the government, NGOs, and international organizations for many years. Some of the **initiatives taken by the Indian government** to improve WASH include:

1. ***Swachh Bharat Abhiyan*** (Clean India Campaign) is a nationwide campaign launched in 2014 to eliminate open defecation, improve solid waste management, and promote hygiene and cleanliness. The campaign has been successful in increasing access to toilets and reducing open defecation.

2. ***Jal Jeevan Mission***, a flagship program launched in 2019 with the aim of providing tap water connections to every rural household by 2024. The program aims to provide safe and adequate drinking water to all households in rural areas.
3. ***National Rural Drinking Water Programme (NRDW)*** launched in 2009 with the aim of providing safe and adequate drinking water to rural areas. The program focuses on creating sustainable drinking water sources, promoting water conservation, and improving water quality.
4. ***National Urban Sanitation Policy (NUSP)*** launched in 2008 with the aim of promoting sanitation and hygiene in urban areas. The policy focuses on creating sustainable sanitation infrastructure, promoting behaviour change, and improving waste management.

3.4.2 Strategies for water provisioning and management

Water provisioning and management are undertaken through a combination of policies, regulations, and practices that aim to ensure the sustainable use and distribution of water resources.

Some of the **key strategies** employed are as follows:

1. ***Water conservation and efficiency measures***: These include promoting water-efficient technologies and practices, such as low-flow fixtures, drought-resistant landscaping, and water reuse.
2. ***Water pricing and incentives*****:** Pricing mechanisms and incentives can help to encourage more sustainable water use and conservation practices.
3. ***Investments in water infrastructure****:* Investment in water infrastructure such as dams, reservoirs, and treatment plants can improve access to water and sanitation services.
4. ***Integrated water resources management***: This approach aims to manage water resources in a holistic and integrated manner, taking into account the needs of all stakeholders and balancing social, economic, and environmental considerations.
5. ***Water governance and institutional arrangements***: Effective water governance and institutional arrangements are critical for ensuring the equitable distribution and sustainable management of water resources.
6. ***International cooperation****:* Global collaboration is critical for addressing transboundary water issues and promoting sustainable water management practices worldwide.

3.5 TELECOMMUNICATION

Telecommunication is defined as the transmission of information – voice, data and multimedia, over electronic media across large distances. It uses various types of transmission technologies, such as, over wire, electromagnetic, radio and optical, and requires infrastructure for radio and television broadcasting, wired and wireless devices, fibre optic cables, satellites, and networks etc. Telecommunication infrastructure is a key driver of economic development, enabling access to information, markets, and resources that can help to spur innovation and growth. The importance of telecommunication infrastructure can be seen in several domains, such as :

1. ***Business and commerce***: Telecommunication infrastructure is essential for conducting business and commerce across distances, facilitating transactions, and enabling access to markets and customers around the world.
2. ***Education and research*:** Telecommunication infrastructure is increasingly important in education and research, enabling distance learning, remote collaboration, and access to online resources.
3. ***Public safety and emergency response***: Telecommunication infrastructure is critical for public safety and emergency response, enabling communication between emergency responders and the public during times of crisis.
4. ***Social connectivity and cultural exchange***: Telecommunication infrastructure provides a means for people to connect with each other, regardless of their location, enabling social connectivity and cultural exchange.

3.5.1 *Telecom infrastructure*

Telecommunication infrastructure refers to the physical networks, equipment, and facilities used to transmit voice, data, and multimedia communication over long distances. There are several types of telecommunication infrastructure, broadly for wired and wireless communication. Wired networks use physical cables, such as copper or fibre optic cables, to transmit data over long distances, as in, traditional landline telephone networks, cable television networks, and high-speed internet networks. Whereas Wireless networks use radio waves to transmit data over the air, without the need for physical cables, such as, cellular networks, satellite networks, and Wi-Fi networks.

However, both require physical facilities and equipment, such as;

- **Telecom towers or cell towers** are structures used to facilitate wireless communication between mobile devices and the telecommunications network. These towers are typically tall structures, ranging from 30 meters to over 100 meters in height, and are equipped with antennas, transmitters, and receivers that enable wireless communication. They are typically located in urban and rural areas, along highways and major roads, and in remote areas where coverage is limited, and maybe freestanding or mounted on existing structures, such as buildings or utility poles. They can also be designed to accommodate multiple carriers, allowing different providers to share the same tower and reduce costs. However, the concern of exposure to electromagnetic radiation and its impact on human health is much debated.
- **Cables** can be installed over ground, underground or under seas. Overground cables are typically strung between poles, buildings, or other structures, and are commonly used in urban and suburban areas, where the cost and complexity of underground installation can be prohibitive. Overground cables are typically made of copper or fibre optic materials, and are designed to withstand weather and environmental conditions. Underground

cables are buried beneath the ground either direct-buried or installed in conduits or ducts, and are commonly used in urban areas. While these are more expensive and difficult to install, underground cables are more reliable and secure, however other activities such as road construction or excavations and servicing sewers, etc can lead to damaging of underground cables. Undersea cables are physical cables laid on the ocean floor to connect different continents and regions, in line with underground but the environmental conditions and installation is extremely challenging.

- **Other Transmission equipment** and devices, i.e., the equipment used to transmit and receive signals, such as, antennas, modems, routers, and switches.
- **Internet exchange points (IXPs)** are physical locations where different internet service providers (ISPs) connect their networks to exchange data. IXPs are essential for enabling the flow of data across different networks and supporting the growth of the internet.
- **Data centers** are facilities used to store, manage, and process large amounts of data, particularly in the areas of cloud computing and internet-based services.

3.5.2 Present and future Challenges in Telecom

Telecommunication infrastructure faces a range of challenges that can impact its effectiveness and ability to support modern communication needs, some of which are as follows:

1. ***Access and Connectivity gaps:*** Despite significant progress in expanding telecommunication infrastructure, there are still areas around the world that lack access to reliable and high-speed internet connectivity, particularly in rural and remote areas. Addressing these gaps in connectivity requires significant investment and infrastructure development.
2. ***Cybersecurity threats:*** Telecommunication infrastructure is vulnerable to a range of cybersecurity threats, including hacking, data breaches, and cyber-attacks. Ensuring the security of telecommunication networks and devices is critical to protecting sensitive information and maintaining public safety.
3. ***Cost and affordability:*** Telecommunication infrastructure can be expensive to build and maintain, which can impact its availability and affordability, particularly in developing countries. Ensuring that telecommunication services are accessible and affordable to all is essential for supporting economic development and social connectivity.
4. ***Regulation and policy***: Telecommunication infrastructure is subject to a range of regulatory and policy frameworks that can impact its development and implementation. Ensuring that policies and regulations support the growth and effectiveness of telecommunication infrastructure can be challenging, particularly in rapidly changing technology environments.
5. ***Integrating Emerging technologies****:* The telecommunication industry is constantly evolving, with new technologies and integrating these emerging technologies into existing infrastructure can be challenging, requiring significant investment and expertise.

The India Story

India is the second-largest telecommunications market globally, with a consistently growing subscriber base and broadband subscriptions. As of December 2022, the tele-density reached 84.56%, while broadband subscriptions reached 832.2 million. The total subscriber base stood at 1170.38 million. In the first quarter of FY23, the telecom sector's gross revenue was Rs. 76,408 Crores, i.e., US$ 9.3 billion (IBEF).

In June-September 2022, the total number of internet subscribers reached 850.95 million, with the wireless segment contributing 95.4% of the total telephone subscriptions. Among the different data technologies, 2G accounted for 0.16%, 3G for 1.02%, and 4G for 98.81% of the total wireless data usage. The rise in mobile-phone penetration and declining data costs are expected to bring 500 million new internet users to India in the next five years, creating opportunities for new businesses. Government initiatives such as the BharatNet Project Scheme, Telecom Development Plan, Aspirational District Scheme, and Comprehensive Telecom Development Plan (CTDP) in the North-Eastern Region have led to a significant growth of 200% in rural internet subscriptions from 2015 to 2021. The integration of payments on unified payments interface (UPI) enabled higher and easier usage. Department of Telecommunication (DoT) launched '*Tarang Sanchar'* - a web portal sharing information on mobile towers and EMF Emission Compliances.

In the past four years, there has been over 75% increase in internet coverage, from 251 million users to 446 million, and it is estimated that by 2025, India will require approximately 22 million skilled workers in 5G-centric technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), robotics, and cloud computing.

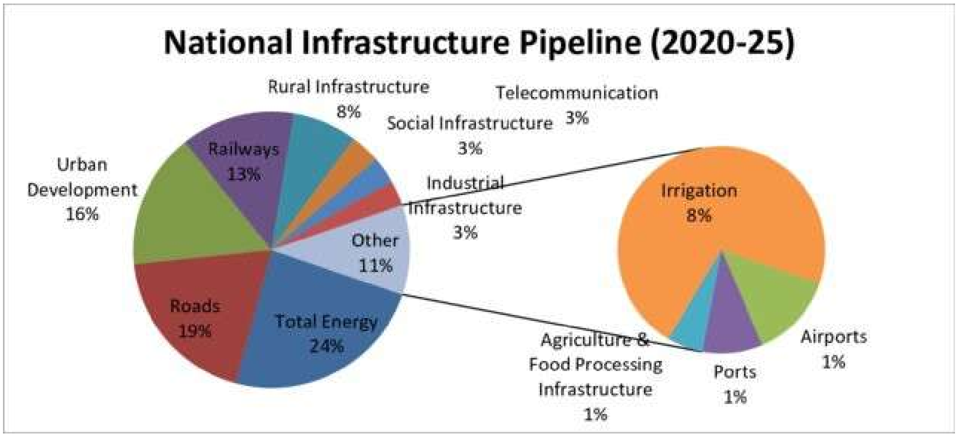

Fig. 3.8: National Infrastructure Pipeline, 2022-25, Sector-wise Fund allocation in % (source : DoEA, MoF, 2019)

3.6 INFRASTRUCTURE DEVELOPMENT STANDARDS & CODES

The **World Standards cooperation (WSC)**, a collaboration between; International Organization for Standardization (ISO), International Telecommunication Union (ITU), and International Electrotechnical Commission (IEC), sets the global framework for standards for many sectors including infrastructure and forms the backbone for the national standards in India. In the recent Global Quality Infrastructure Index (GQII) 2021, which ranks 184 economies in the world on the basis of the quality infrastructure (QI), **India's national accreditation system** under the Quality Council of India (QCI) has *ranked 5th in the world*, and our overall QI system ranked at the 10th position, with the standardization system (under BIS) at 9th and the metrology system (under NPL-CSIR) at 21st position in the world. The Bureau of Indian Standards (BIS), a statutory body established through legislation through The Bureau of Indian Standards Act 1986, adopts from ISO and IEC standards while the Ministry of Communication and Information Technology have ITU membership. The other standards making bodies of India, apart from BIS are; Telecom Engineering Centre (TEC), Telecommunications Standards Development Society of India (TSDSI), Automotive Research Association of India (ARAI), and Research Designs and Standards organization (RDSO).

In India, the roles and responsibilities for development of infrastructure is defined by the Constitution of India. While the centre is responsible for the critical national-level infrastructures, such as, National highways, Railways, Major ports, Airports and Telecom, the State together with the Municipality and Panchayats are responsible for regulating building construction, water management and supply, urban town planning, other roads and bridges, rural housing and rural electricity. To aid in Public-Private Partnership to see through infrastructural development, in 2007 the Centre set up the **India Infrastructure Project Development Fund (IIPDF) Scheme** aimed at creating appropriate mechanisms, guidelines, advisories, and funding support.

3.6.1 International and National Codes for Civil Engineering and Construction

There are a large number of **Indian Standard (IS) codes** that support the practice of civil engineering and architecture for safe and durable infrastructure construction. These offer guidelines, specifications, and safety prerequisites for construction materials, design parameters, testing techniques, and construction practices, and serve as a reference to guarantee consistency, safety, and excellence in civil engineering projects throughout the country.

Civil engineers refer to these codes for design and analysis of structures, as well as for specifications, methods and code of practice, for e.g., IS : 456; 10262; Sp 23 provides 'codes for designing concrete mixes', while IS : 2386 provides 'methods for tests for aggregate for concrete' and IS : 4925 provides 'specifications for concrete batching plant'. Dedicated list of standards is available for materials and elements, such as, Cement & Concrete, which in turn has codes on cement (IS 269, IS 8041, IS 650), coarse / fine Aggregate (IS 383, IS 2386), Masonry Mortar, Cement Concrete, Curing Compound, etc. Other codes cover Lime and Gypsum, and Doors, Windows and Shutters.

Nationally recognised, some of the **IS codes on civil engineering** are;

Code No.	Description
IS - 4031	Method of physical tests for hydraulic Cement
IS - 650	Specification for Standard sand for testing of Cement
IS - 383	Specification for Coarse and Fine aggregate for use in mass concrete
IS - 515	Specification for natural and manufactured aggregate for use in mass concrete.
IS - 2387	Method of test for aggregates for concrete.
IS - 516	Methods of test for strength of concrete.
IS - 1199	Methods of sampling and analysis of concrete
IS - 3025	Methods of sampling and test (physical and chemical) for water used in industry.
IS - 432	Specification for Mild steel and medium tensile bars and hard drawn steel wire.
IS - 1139	Specification for hot rolled mild steel, medium tensile steel and high yield strength steel deformed bars for concrete reinforcement.
IS – 1566	Specification for plain hard drawn steel wire fabric for concrete reinforcement
IS - 1785	Specification for plain hard drawn steel wire for prestressed concrete.
IS - 1786	Specification for cold twisted steel high strength deformed bars for concrete reinforcement.
IS - 303	Specification for Plywood for general purposes

Internationally recognised, the **ISO ICS 93 codes** are on civil engineering, and covers information on the following;

ISO Code No.	Description
93.010	Civil engineering in general Construction drawings, see 01.100.30
93.020	Earthworks. Excavations. Foundation construction. Underground works Including geotechnics. Earth-moving machinery, see 53.100
93.025	External water conveyance systems Including buried and above ground installations Pipelines and its parts for external water conveyance systems, see 23.040.03 Internal water supply systems, see 91.140.60
93.030	External sewage systems Sewage water disposal and treatment, see 13.060.30 Pipelines and its parts for external sewage systems, see 23.040.05 Internal drainage systems, see 91.140.80
93.040	Bridge construction
93.060	Tunnel construction
93.080	Road engineering
93.100	Construction of railways Including the construction of tramways, funicular railways, cableways, rail traffic control equipment and installations, etc. Rails and railway components, including track, see 45.080 Equipment for railway/cableway construction and maintenance, see 45.120
93.110	Construction of ropeways Ropeway equipment, see 45.100 Equipment for ropeway construction and maintenance, see 45.120
93.120	Construction of airports Including air transport control equipment and installations
93.140	Construction of waterways, ports and dykes Including river embankments, water transport control equipment and installations, etc.
93.160	Hydraulic construction Hydraulic energy equipment, see 27.140

In addition, the **National Building Code (NBC),** discussed in detail in Unit 5**,** a model code for adoption of across the nation and contains administrative regulations, development control rules and general building requirements; fire safety requirements; stipulations regarding materials, structural design and construction (including safety); construction management practices and safety, building and plumbing services; approach to sustainability; and asset and facility management is referred to during design and detailing construction working drawings .

Eventually the quality of design and execution is the onus of the Civil engineer and Architect. Thus, to aid in other complimentary areas, Niti Ayog has compiled the '**Indian Infrastructure Body of Knowledge –** A technical Baseline to the Practice of Program and Project Management in India' under the National Program and Project Management Policy Framework with the intention of "*laying down a plan of action and advocating short-term and long-term strategies for improving Program and Project Management practices in India, as well as align with the global best practices*" (CEO, Niti Ayog)

3.7 INNOVATIONS AND METHODOLOGIES FOR SUSTAINABILITY

Sustainable Infrastructure, be it built, natural or hybrid, are systems that are "*planned, designed, constructed, operated and decommissioned in a manner that ensures economic and financial, social, environmental, including climate resilience, and institutional sustainability over the entire infrastructure life cycle*" (UNEP). The Organisation for Economic Co-operation and Development (OECD) estimates that an annual investment of USD 6.9 trillion is needed for infrastructure to meet development goals and create a low carbon, climate resilient future by 2050. OECD's ***Strategic Policies for Sustainable Infrastructure*** identifies various themes that require attention, such as, Low-carbon transition, Technology and Innovation, Inclusiveness and Accessibility, etc.

Presently the OECD is developing a toolkit on quality infrastructure investment for policymakers and practitioners, based on the ***G20 Principles for Quality Infrastructure Investment (QII)***, developed under the Japanese G20 Presidency, that stated, "*quality infrastructure investment contributes to maximizing the positive impact of infrastructure to achieve sustainable growth and development, raising economic efficiency in view of life cycle costs, integrating environmental and social considerations in infrastructure, building resilience against natural disasters, and strengthening infrastructure governance*."

The UN Environment Assembly (UNEA) Members States in March 2023 adopted a resolution on '***Sustainable and Resilient Infrastructure'*** encouraging them to;

- provide opportunities for engaging relevant stakeholders,
- promote investment in sustainable and resilient infrastructure, natural infrastructure and nature-based solutions,
- cooperate internationally to strengthen frameworks, including for financing, and

- implement the '*International Good Practice Principles for Sustainable Infrastructure'*, which in turn has the following guiding principles ;
    1. Strategic Planning
    2. Responsive, Resilient, And Flexible Service Provision
    3. Comprehensive Life Cycle Assessment of Sustainability
    4. Avoiding Environmental Impacts and Investing In Nature
    5. Resource Efficiency and Circularity
    6. Equity, Inclusiveness, And Empowerment
    7. Enhancing Economic Benefits
    8. Fiscal Sustainability and Innovative Financing
    9. Transparent, Inclusive, And Participatory Decision-Making
    10. Evidence-Based Decision-Making

There are several **innovations and methodologies** that can be employed to ensure the sustainability of infrastructure development, such as :

1. ***Green infrastructure*** is a concept of incorporating the importance of Environment and considering the impact of decisions on it while developing infrastructure strategies. It involves the use of natural systems and materials to provide sustainable solutions, for e.g., green roofs, permeable pavements, and rain gardens, which can help to reduce stormwater runoff and mitigate the urban heat island effect.
2. ***Integrated Water Resources Management (IWRM)*** is a process that promotes the "*coordinated development and management of water, land and related resources in order to maximise economic and social welfare in an equitable manner without compromising the sustainability of vital ecosystems*" (UNEP).
3. ***Circular Economy*** or circularity is an economic model of production and consumption that seeks to eliminate waste and promote the sustainable use of resources. In the context of infrastructure development, it can be used to design infrastructure projects that prioritize the use of renewable materials, reduce waste, and promote the reuse and recycling of materials.
4. ***Tool and methods for reducing Environmental Impact***, such as, Environmental Impact Assessment (EIA) is a methodology used "*to identify the environmental, social and economic impacts of a project prior to decision-making*, Life-Cycle Assessment (LCA) is a tool used to "*evaluate the environmental impact of infrastructure projects throughout their entire life cycle, from raw material extraction to disposal*", and Building Information Modelling (BIM) is a digital modelling technology that enables the creation of detailed 3D models of infrastructure projects, to help improve project efficiency, reduce waste, and optimize the use of materials and resources.

Infrastructure design and development is crucial for a Nation's growth and civil engineers play the most pivotal role in ensuring quality, safety, innovation, and sustainability of the same, bearing great societal and global impact.

UNIT SUMMARY

This unit focusses on present and future projection of various facets of Infrastructure development by discussing concepts such as 'Smart City' and further delves into understanding critical infrastructure, such as, Transportation, Energy, Water resources management and Telecommunication. It also offers an overview on the Standards and Codes relevant for infrastructure and construction industry, and culminates with a discussion on present day initiatives, innovations and methodologies employed in this sector to ensure sustainability.

EXERCISES

I. Multiple Choice Questions

Q. 3.1 Which of the following are 'critical infrastructure' as per India ?
- (a) Disaster Management
- (b) Aviation
- (c) Cyber security
- (d) all of the above

Q. 3.2 What is the population of a 'Megacity' ?
- (a) 1 to 5 million
- (b) over 10 million
- (c) less than 10 million
- (d) 1 million

Q. 3.3 Which ancient Indian University town, now in present day Afghanistan, was connected to Prayagraj by the Great Road built by Mauryan Empire?
- (a) Bhagalpur
- (b) Gandhar
- (c) Takshashila
- (d) Benaras

Q. 3.4 What is the world's longest road tunnel?
(a) the Lærdalstunnelen
(b) Channel Tunnel
(c) Yamate Tunnel
(d) Gotthard Base Tunnel

Q. 3.5 Which of the following renewable energy source may be argued to not be a 'clean energy' source?
(a) Solar
(b) Biomass
(c) Geothermal
(d) Wind

Answers of Multiple Choice Questions: 3.1 (d) , 3.2 (b) , 3.3 (c), 3.4 (a), 3.5 (b

II. Short and Long Answer Type Questions

Q. 3.6 What is a 'Smart City'? Briefly explains the characteristics of a Smart City.

Q. 3.7 What are some of the potential new sources of energy? What are the associated challenges and impacts?

Q. 3.8 Define WASH? What are the various initiatives by Govt. of India to promote WASH?

Q. 3.9 What are the various types of physical facilities and equipment required as part of Telecommunication infrastructure? Discuss the challenges that impact Telecommunication infrastructure?

Q. 3.10 What is the importance of Infrastructure development standards and codes? Illustrate with examples how this supports the profession of civil engineering.

KNOW MORE

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